

Hereditary Sensory and Autonomic Neuropathy Type 1A (HSAN1A): Comprehensive Disease Characterization Report

Disease: Hereditary Sensory and Autonomic Neuropathy Type 1A **MONDO ID:** MONDO:0008086 | **OMIM:** 162400 | **Gene:** *SPTLC1* (9q22.31) **Category:** Mendelian (autosomal dominant)

Summary

Hereditary Sensory and Autonomic Neuropathy Type 1A (HSAN1A) is the most common autosomal-dominant hereditary sensory neuropathy. It is caused by heterozygous, mostly missense mutations in *SPTLC1*, which encodes long-chain base subunit 1 of **serine palmitoyltransferase (SPT)**, the rate-limiting enzyme of de novo sphingolipid biosynthesis. Rather than acting through simple loss of function, HSAN1A mutations are **neomorphic gain-of-function** changes: they shift the amino-acid substrate preference of SPT away from its canonical substrate L-serine toward L-alanine (and glycine). This produces atypical **1-deoxysphingolipids (1-deoxySLs)** that lack the C1 hydroxyl group, cannot be converted into complex sphingolipids or degraded by the normal catabolic machinery, accumulate to neurotoxic levels, and preferentially injure long dorsal-root-ganglion (DRG) sensory neurons ([PMID: 11242114](#); [PMID: 29778900](#); [PMID: 27016021](#)).

Clinically, HSAN1A presents in the second-to-third decade with length-dependent, sensory-predominant axonal neuropathy: distal lower-limb sensory loss (pain and temperature prominent), spontaneous shooting/lancinating pain, and variable distal motor involvement. Progressive loss of protective sensation leads to the hallmark **ulceromutilating complications** — recurrent painless foot ulcers, osteomyelitis, neuropathic (Charcot) arthropathy, and distal amputations. The disease shows incomplete penetrance, variable expressivity, and apparent

anticipation ([PMID: 16364956](#); [PMID: 11242114](#)). The canonical mutation p.Cys133Trp (c.399T>G) is a **southern-English founder allele** and the single most common cause of HSAN in the UK ([PMID: 11479835](#); [PMID: 22302274](#)).

The mechanistic understanding of HSAN1A has direct therapeutic consequences. Because the disease is driven by a substrate-availability defect, **high-dose oral L-serine** competitively restores the canonical reaction, lowers plasma 1-deoxySLs, and has Class I randomized-controlled-trial evidence for slowing progression — the first disease-modifying therapy for the condition ([PMID: 30626650](#); [PMID: 22045570](#)). Cryo-EM structures of the human SPT–ssSPT–ORMDL complex have resolved how catalytic-pocket variants alter substrate selectivity and, importantly, revealed that a distinct set of *SPTLC1* transmembrane-domain variants instead impair ORMDL/ceramide feedback and cause juvenile ALS — establishing that treatment must be genotype-guided ([PMID: 33558762](#); [PMID: 37308477](#); [PMID: 35900868](#)). The same 1-deoxySL/serine-deficiency biology also links HSAN1A to retinal macular telangiectasia type 2, revealing a shared neuronal-retinal degenerative mechanism ([PMID: 31509666](#)).

1. Disease Information

Overview. HSAN1A is a rare, adult-onset, autosomal-dominant, length-dependent axonal peripheral neuropathy in which sensory fibers are predominantly affected, with variable motor and autonomic involvement. It belongs to the broader group of **hereditary sensory neuropathies (HSNs)** / hereditary sensory and autonomic neuropathies (HSANs), also historically classified among the ulceromutilating neuropathies and, when motor weakness is prominent, overlapping with Charcot-Marie-Tooth (CMT) disorders ([PMID: 15319794](#); [PMID: 12633143](#)).

Key identifiers.

Resource	Identifier
MONDO	MONDO:0008086
OMIM	162400 (HSAN1A)
Gene	<i>SPTLC1</i> , OMIM 605712, HGNC:11277, 9q22.31
Orphanet	Hereditary sensory and autonomic neuropathy type 1 (ORPHA:36386)
ICD-10	G60.8 (other hereditary and idiopathic neuropathies)
ICD-11	8C20 (hereditary neuropathy) range
MeSH	Hereditary Sensory and Autonomic Neuropathies (D009477 family)

Synonyms / alternative names. HSAN I; HSAN1; HSN1; HSN I; Hereditary Sensory Neuropathy Type 1; HSN type IA; Neuropathy, Hereditary Sensory Radicular, Autosomal Dominant; ulceromutilating neuropathy (historical).

Source of information. This report is derived from **aggregated disease-level resources** — OMIM, Orphanet, and primary/peer-reviewed literature (family cohorts, case series, functional studies, cryo-EM structures, and one randomized controlled trial) — rather than from individual electronic health records.

2. Etiology

Primary cause — genetic. HSAN1A is caused by heterozygous, autosomal-dominant mutations in *SPTLC1*. The genetic mechanism is a **neomorphic gain of function / dominant change of substrate specificity**, not haploinsufficiency. Canonical mutations (p.Cys133Trp, p.Cys133Tyr, p.Val144Asp, and the p.Ser331 variants) alter SPT so that it condenses palmitoyl-CoA with L-alanine and glycine instead of L-serine, generating 1-deoxysphinganine and 1-deoxymethylsphinganine. As Bode & Bode (2018) summarize: *"Mutations in SPT result in a change in enzyme substrate specificity, which causes the production of atypical deoxysphinganine and deoxymethylsphinganine, rather than the normal enzyme product, sphinganine. Levels of these abnormal compounds are elevated in blood of HSN-1 patients"* (PMID: 29778900; PMID: 11242114).

Genetic risk factors. - **Causal variants:** *SPTLC1* missense variants (see Section 4). Allelic disorders also arise from *SPTLC2* (HSAN1C), which likewise raises 1-deoxySLs (PMID: 26573920). - **Modifier genes:** SPT small subunits (SPTSSA/SPTSSB; yeast ortholog Tsc3) regulate serine-versus-alanine selectivity and are candidate modifiers of 1-deoxySL output (PMID: 30154231; PMID: 33558762).

Environmental / metabolic risk factors. Because the deleterious reaction depends on relative amino-acid availability, **low systemic L-serine and high L-alanine** shift the balance toward 1-deoxySL formation. In C133W transgenic mice an L-alanine-enriched diet increased deoxySLs and worsened neuropathy (PMID: 22045570). Metabolic states that lower serine or raise alanine (and, more broadly, conditions raising 1-deoxySLs such as diabetes/metabolic syndrome) are plausible aggravating factors. Sex may modulate motor severity (earlier/more severe motor onset reported in males in some families; PMID: 16364956).

Protective factors. **High dietary/therapeutic L-serine** is the key protective factor — it competes L-alanine out of the SPT active site, reducing 1-deoxySL synthesis (PMID: 22045570; PMID: 30626650). No specific protective germline alleles are established.

Gene–environment interaction. The clearest GxE axis is **genotype × serine/alanine availability**: the pathogenic genotype only produces toxic lipids in the context of available L-alanine, and disease burden is tunable by dietary amino-acid manipulation in both directions — a rare example of a Mendelian neuropathy with a dietary modifier that has been exploited therapeutically.

3. Phenotypes

HSAN1A phenotypes are predominantly **sensory clinical signs and symptoms**, with secondary physical manifestations from denervation.

Phenotype	Type	Onset	Severity/ Progression	Frequency	Suggested HPO
Distal sensory loss (lower limbs, length-dependent)	Clinical sign/symptom	2nd–3rd decade	Progressive	Near-universal (core feature)	HP:0106487 (impaired distal vibration); HP:0007328 (impaired pain sensation)
Loss of pain & temperature sensation	Symptom	Early	Progressive	Prominent	HP:0007328; HP:0010829 (impaired temperature sensation)
Spontaneous shooting/lancinating (neuropathic) pain	Symptom	Variable	Episodic/fluctuating	Common	HP:0009830 (peripheral neuropathy); HP:0000738
Distal foot ulcers (painless)	Physical manifestation	After sensory loss	Recurrent, progressive	Frequent/feared complication	HP:0200042 (skin ulcer)
Osteomyelitis / bone infection	Physical manifestation	Later	Progressive	Frequent complication	HP:0002754 (osteomyelitis)
Acro-osteolysis / distal amputations	Physical manifestation	Later	Mutilating	Frequent	HP:0012060 (acro-osteolysis)
Neuropathic (Charcot) arthropathy	Physical manifestation	Later	Progressive	Variable	HP:0100261 (neuropathic joint)
Distal muscle wasting/weakness	Clinical sign	After sensory onset	Variable; sometimes early/severe	Variable	HP:0003693 (distal amyotrophy); HP:0002460

Phenotype	Type	Onset	Severity/ Progression	Frequency	Suggested HPO
Autonomic involvement (e.g., distal hypohidrosis)	Symptom	Variable	Variable	Variable/ mild	HP:0000966 (hypohidrosis)
Sensorineural hearing loss (variant families)	Clinical sign	Adult	Variable	Subset	HP:0000407

Onset, severity, progression, frequency. Onset is typically the second-to-third decade, with initial sensory loss in the feet followed by distal muscle wasting and weakness; as the original gene-identification paper states, *"Initial symptoms are sensory loss in the feet followed by distal muscle wasting and weakness. Loss of pain sensation leads to chronic skin ulcers and distal amputations"* (PMID: 11242114). The course is chronic and slowly progressive, with reduced penetrance in some carriers and earlier onset in younger generations (anticipation) (PMID: 16364956).

Quality-of-life impact. Insensate feet, recurrent ulceration, osteomyelitis, chronic neuropathic pain, and amputations produce substantial disability affecting mobility, employment, and self-care. Disease-specific QoL instruments are not established; the CMT Neuropathy Score (CMTNS) is used as a composite disability/severity outcome (PMID: 30626650).

4. Genetic / Molecular Information

Causal gene. *SPTLC1* (serine palmitoyltransferase long chain base subunit 1), 9q22.31, OMIM 605712, HGNC:11277. It encodes subunit 1 of the SPT heterodimer (with SPTLC2/SPTLC3), the rate-limiting, PLP-dependent enzyme of de novo sphingolipid synthesis. Dawkins et al. mapped *SPTLC1* to chromosome 9q22.1–22.3 and confirmed it as the HSN1 gene: *"We found two mutations to be located in exon 5 (C133Y and C133W) and one mutation to be located in exon 6 of SPTLC1 (V144D). All families showing definite or probable linkage to chromosome 9 had mutations in these two exons"* (PMID: 11242114).

Pathogenic variants (HSAN1A).

Variant (protein)	cDNA	Exon	Class	Notes
p.Cys133Trp	c.399T>G	5	Pathogenic	Most common; southern-English founder allele
p.Cys133Tyr	—	5	Pathogenic	Original cohort
p.Val144Asp	—	6	Pathogenic	Original cohort
p.Ser331Phe / p.Ser331Tyr	—	—	Pathogenic	"Ser331 syndrome" — severe, HSAN1A + motor neuron disease overlap

- **Classification:** Established *SPTLC1* HSAN1 variants are ACMG **Pathogenic/Likely Pathogenic** (functional data + segregation + absence from population databases).
- **Variant type/class:** Predominantly **missense**; disease acts by altered substrate specificity, not truncation.
- **Allele frequency:** Pathogenic alleles are essentially absent from gnomAD/population databases.
- **Origin:** **Germline**, autosomal dominant. Parental **mosaicism** has been documented for the allelic ALS branch (asymptomatic parent with ~17% mosaic p.Ala20Thr; [PMID: 39666121](#)).
- **Functional consequence:** **Neomorphic gain of function** — shift to L-alanine/glycine usage → 1-deoxySL production ([PMID: 29778900](#)).

Allelic branch (distinct disease). *SPTLC1* transmembrane-domain variants (p.Ala20Ser, p.Ala20Thr, p.Leu38Arg, p.Ser331Tyr) impair binding of ORMDL (negative regulators of SPT), releasing feedback inhibition and increasing overall canonical sphingolipid synthesis (sphinganines/ceramides) → **juvenile/childhood-onset ALS** rather than HSAN1A. Johnson et al. report "*De novo variants in SPTLC1 (p.Ala20Ser in 2 patients and p.Ser331Tyr in 1 patient)*" causing juvenile ALS ([PMID: 34459874](#)), and the mechanism is that "*SPTLC1-ALS variants map to a transmembrane domain that interacts with ORMDL proteins... ORMDL binding to the holoenzyme complex is impaired in cells expressing pathogenic SPTLC1-ALS alleles, resulting in increased SL synthesis and a distinct lipid signature*" ([PMID: 35900868](#); [PMID: 37348646](#)). The p.Ser331 variants uniquely show "*the coexistence of neurotoxic deoxy-sphingolipids with an excess of canonical products of the SPT enzyme*" ([PMID: 35904184](#); [PMID: 36964315](#)).

Modifier genes. SPTSSA/SPTSSB (small SPT subunits; yeast Tsc3) tune serine-versus-alanine choice and may modify deoxySL output; "*hereditary sensory and autonomic neuropathy type I results from SPT mutations that cause an abnormal accumulation of alanine-derived SPLs. The regulatory mechanism for SPT amino acid selectivity ... [is] unknown*" ([PMID: 30154231](#)).

Epigenetic / chromosomal. No specific epigenetic signature or chromosomal abnormality is implicated; the disorder is a single-gene point-mutation disease.

5. Environmental Information

- **Environmental / dietary factors:** The dominant non-genetic modifier is **amino-acid availability** — L-serine (protective) versus L-alanine (aggravating), demonstrated by diet manipulation in transgenic mice ([PMID: 22045570](#)).
 - **Chemical/toxin relevance:** 1-DeoxySLs also mediate **chemotherapy-induced peripheral neuropathy** (paclitaxel, docetaxel), where SPT uses L-alanine and 1-deoxySLs accumulate in DRG — a convergent mechanism relevant to disease modeling but not a cause of HSAN1A itself ([PMID: 32058598](#)).
 - **Lifestyle factors:** No established smoking/alcohol/exercise associations specific to HSAN1A.
 - **Infectious agents:** Not applicable — HSAN1A is a monogenic disorder. (Secondary wound infections/osteomyelitis are complications, not causes.)
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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A heterozygous **missense mutation in *SPTLC1*** (e.g., p.Cys133Trp) alters an active-site–proximal residue of the SPT catalytic core → **leads to** a change in the enzyme's amino-acid substrate selectivity.
2. Mutant SPT **condenses palmitoyl-CoA with L-alanine and glycine** instead of L-serine → **results in** synthesis of atypical **1-deoxysphinganine and 1-deoxymethylsphinganine (1-deoxySLs)**. As stated for HSAN1: *"1-Deoxysphingolipids (1-deoxySL) are atypical and neurotoxic sphingolipids formed by alternate substrate usage of the enzyme serine-palmitoyltransferase. Pathologically increased 1-deoxySL formation causes hereditary sensory and autosomal neuropathy type 1 (HSAN1)"* ([PMID: 27016021](#)).
3. Because 1-deoxySLs **lack the C1 hydroxyl**, they **cannot** be converted to complex sphingolipids or enter the canonical degradation pathway (S1P-lyase route) → **leads to** their

intracellular and circulating **accumulation** (elevated in patient plasma; used as a biomarker).

4. Accumulated 1-deoxySLs are **neurotoxic** to sensory neurons → **cause** (branch):
5. 4a. **Ca²⁺ handling abnormalities** and **mitochondrial dysfunction** (PMID: 29778900);
6. 4b. **Cytoskeletal disruption** (neurite/axon swelling, retraction);
7. 4c. **NMDA-receptor-mediated excitotoxic signaling** (PMID: 27016021).
8. These converge on **dose- and time-dependent reduction of neurite length and neuronal survival** → **results in** degeneration of long **dorsal-root-ganglion (DRG) sensory neurons** and their axons (length-dependent, distal-first).
9. Distal sensory axonopathy **leads to** loss of protective pain/temperature sensation, neuropathic pain, and (variably) distal motor axon loss → **results in** the clinical picture of ulceromutilating sensory neuropathy.
10. Loss of protective sensation → **causes** repeated unperceived trauma → **leads to** foot ulcers → **osteomyelitis, Charcot arthropathy, and distal amputations** (downstream, secondary tissue-injury cascade).

Branch (allelic, not HSN1A): Transmembrane *SPTLC1* variants **impair ORMDL/ceramide feedback** → **increase canonical sphingolipid synthesis** → **juvenile ALS/motor neuron degeneration**; structurally, "*childhood amyotrophic lateral sclerosis (ALS) variants in the SPTLC1 subunit cause impaired ceramide sensing in the SPT-ORMDL3 mutants*" (PMID: 35900868; PMID: 37308477).

Detail by category

- **Molecular pathways / biochemistry:** De novo sphingolipid biosynthesis (SPT is the rate-limiting, PLP-dependent condensation step). The defect is a **substrate-specificity shift** producing 1-deoxySLs. Upstream driver = mutation; downstream = lipotoxicity.
- **Cellular processes:** Mitochondrial dysfunction, disrupted Ca²⁺ homeostasis, cytoskeletal/axonal transport failure, and excitotoxicity; net effect is neuronal apoptosis/axon degeneration (PMID: 29778900; PMID: 27016021).
- **Protein dysfunction:** The mutant SPT holoenzyme is not simply inactivated — it is functionally **reprogrammed** (neomorphic). Cryo-EM shows "*SPTLC1 and SPTLC2 form a dimer of heterodimers as the catalytic core. SPTssa participates in acyl-CoA coordination, thereby stimulating the SPT activity and regulating the substrate selectivity. ORMDL3 is located in the center of the complex, serving to stabilize the SPT assembly*" (PMID: 33558762).

- **Metabolic changes:** Diversion of sphingoid-base synthesis toward 1-deoxy species; sensitivity to serine/alanine ratio.
- **Molecular profiling:** iPSC-derived HSN1 neurons show L-serine-responsive deficits in neuronal **ganglioside composition** and axoglial interactions ([PMID: 34337561](#)). DRG transcriptomics in an S331F mouse implicate mitochondrial pathways, reversible by allele-specific ASO ([PMID: 41124364](#)).

Suggested ontology terms. GO: sphingolipid biosynthetic process (GO:0030148); serine C-palmitoyltransferase activity (GO:0004758); neuron apoptotic process (GO:0051402); mitochondrion organization (GO:0007005). CL: neuron of dorsal root ganglion / sensory neuron (CL:0000101, CL:0000210). CHEBI: 1-deoxysphinganine; sphinganine; L-serine (CHEBI:17115); L-alanine (CHEBI:16977).

7. Anatomical Structures Affected

- **Organ/body system level: Peripheral nervous system** — primary. Predominantly the **somatosensory** system; variable **motor** and **autonomic** involvement. Secondary: **skin** (ulcers), **bone/joints** (osteomyelitis, acro-osteolysis, Charcot joints), and distal **extremities** (amputations).
 - **Tissue/cell level: Nervous tissue** — dorsal root ganglion **sensory neurons** and their **peripheral axons** (long-fiber, length-dependent). Small (pain/temperature) fibers are prominently affected; myelinated large fibers also lost. Cell Ontology: dorsal root ganglion neuron (CL:0000101), sensory neuron (CL:0000210).
 - **Subcellular level: Mitochondria** (dysfunction), **endoplasmic reticulum** (SPT is an ER-membrane enzyme; site of the primary lesion), **plasma membrane/cytoskeleton**. GO cellular component: endoplasmic reticulum membrane (GO:0005789); serine C-palmitoyltransferase complex (GO:0017059); mitochondrion (GO:0005739).
 - **Localization (UBERON):** dorsal root ganglion (UBERON:0000044); peripheral nerve/sural nerve (UBERON:0001323); sciatic nerve (UBERON:0001322); skin of foot (UBERON:0001513). Distribution is **bilateral and symmetric, distal > proximal, lower limbs > upper limbs**. In allelic MacTel-2 families, the **retina/macula** (UBERON:0000966) is additionally affected.
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8. Temporal Development

- **Onset:** Typically **adult, second-to-third decade; insidious/chronic** onset. Anticipation (earlier onset in successive generations) is reported ([PMID: 16364956](#)). The Ser331 and allelic ALS variants shift onset to childhood/juvenile.
 - **Progression:** Slowly **progressive**, chronic, lifelong. Course begins with distal sensory loss, then distal wasting/weakness and ulceromutilating complications ([PMID: 11242114](#)).
 - **Stages (descriptive):** early (subclinical/sensory symptoms) → intermediate (established sensory loss, neuropathic pain, early ulceration) → advanced (osteomyelitis, Charcot joints, amputations, motor deficits).
 - **Patterns:** No spontaneous remission; deterioration is generally steady. **Critical intervention window:** earlier initiation of 1-deoxySL-lowering therapy (L-serine) before irreversible axon loss is expected to be more beneficial — supported by neonatal-vs-adult efficacy signals in ASO mouse studies ([PMID: 41124364](#)).
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9. Inheritance and Population

- **Inheritance:** Autosomal dominant. Penetrance is incomplete and expressivity variable; apparent **anticipation** occurs. Houlden et al. describe *"lack of penetrance of the SPTLC1 mutation in some individuals, variability in age of onset along with an earlier age of onset in younger generations, in some patients surprisingly early and often severe motor involvement"* ([PMID: 16364956](#)).
- **Epidemiology:** HSAN1 overall is rare (broad HSAN prevalence estimates ~1/100,000–1/1,000,000). In a UK cohort, HSAN gene mutations were found in **14.3% (25/140)** of index patients, and *"The p.Cys133Trp mutation in SPTLC1 is the most common cause of HSAN in the UK population and should be screened first in all patients with sporadic or autosomal dominant HSAN"* ([PMID: 22302274](#)).
- **Founder effects:** The **p.Cys133Trp (c.399T>G)** allele traces to a common **southern-English founder**: *"The Australian and English families may therefore have a common founder who, on the basis of historical information, has been determined to have lived in southern England prior to 1800"* ([PMID: 11479835](#)).
- **Germline mosaicism:** Documented for the allelic ALS branch (parental mosaicism ~17%; [PMID: 39666121](#)).

- **Consanguinity/carrier frequency:** Not relevant for this dominant disorder (unlike recessive HSANs, e.g., HSAN4/*NTRK1*, where consanguinity and founder variants matter — PMID: 41474134).
- **Demographics:** Both sexes affected; some families show **earlier/more severe motor involvement in males** (PMID: 16364956). Geographic clustering follows founder-allele dispersal (UK, Australia, Europe/North America).

10. Diagnostics

Clinical/electrophysiology. - **Nerve conduction studies / EMG:** predominantly **sensory axonal neuropathy** (reduced/absent sensory nerve action potentials), with variable motor axonal features (PMID: 16364956). - **Skin biopsy (PGP9.5):** reduced **intraepidermal nerve-fiber density**; **sural nerve biopsy:** loss of myelinated (and non-myelinated) axons. - **Quantitative sensory testing:** abnormal small-fiber thresholds early.

Biomarker. Elevated **plasma 1-deoxysphingolipids (1-deoxySL)** are a disease-specific biochemical marker and pharmacodynamic readout (they fall with L-serine or ASO therapy) (PMID: 29778900; PMID: 30626650; PMID: 41124364).

Genetic testing. Sequence *SPTLC1* first (especially p.Cys133Trp in UK/founder populations; PMID: 22302274). A **hereditary neuropathy gene panel** (or WES/WGS) covering *SPTLC1*, *SPTLC2*, *RAB7A* (*CMT2B*), *NTRK1*, *IKBKAP/ELP1*, *WNK1/HSN2*, *ATL1*, *DNMT1*, *RFC1*, etc., resolves genetically heterogeneous cases (PMID: 40211677). Single-gene testing suffices when a familial variant is known.

Differential diagnosis. - **HSAN1C** (*SPTLC2*, e.g., p.Arg183Trp): later onset (>50), also raises 1-deoxySLs (PMID: 26573920). - **HSN with cough/GERD** (chromosome 3p22–p24 linked): distinct autonomic/aerodigestive features (PMID: 16311270). - **HSAN4** (*CIPA*, *NTRK1*): congenital, anhidrosis, insensitivity to pain, intellectual disability — recessive (PMID: 41474134; PMID: 41257813). - **Hereditary transthyretin (ATTRv) amyloid polyneuropathy:** sensorimotor + autonomic + cardiac, later onset, amyloid on biopsy (PMID: 23425518). - **CMT2B** (*RAB7A*) and other CMT/HNPP forms with ulceromutilating/overlap features (PMID: 41179108). - **RFC1** (*CANVAS*), **PUM1**, **COX20**-related sensory neuropathies/ataxia (PMID: 40211677).

Screening. Not part of newborn screening. **Cascade genetic testing** of at-risk relatives and predictive testing with genetic counseling are appropriate given AD inheritance.

11. Outcome / Prognosis

- **Survival/mortality:** HSAN1A itself is **not typically life-limiting**; life expectancy is largely preserved. Mortality risk is indirect (sepsis from osteomyelitis, complications of amputation). The allelic **juvenile ALS branch** carries a serious motor-neuron-disease prognosis (though a 30-year survivor is reported — [PMID: 37497262](#)).
 - **Morbidity/disability:** High — chronic neuropathic pain, insensate feet, recurrent ulcers, osteomyelitis, Charcot arthropathy, and amputations drive long-term functional impairment ([PMID: 15319794](#); [PMID: 12633143](#)).
 - **Disease course/complications:** Progressive; complications are largely preventable with meticulous foot care.
 - **Prognostic factors:** Earlier onset, motor involvement (e.g., Ser331 variants), and higher 1-deoxySL burden predict worse course. **Plasma 1-deoxySL** serves as a prognostic/pharmacodynamic biomarker; **CMTNS** tracks disability.
 - **Recovery potential:** Established axon loss is largely irreversible; disease-modifying therapy aims to **slow** progression rather than reverse deficits — motivating early intervention.
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12. Treatment

Disease-modifying — high-dose oral L-serine (first-line, evidence-based). A randomized, placebo-controlled trial (NCT01733407; n=18; **400 mg/kg/day** L-serine for 1 year plus open-label extension) showed *"After 1 year, the l-serine group experienced improvement in CMTNS relative to the placebo group (-1.5 units, 95% CI -2.8 to -0.1)"*, providing **Class I evidence** that L-serine slows progression; it was safe and lowered plasma deoxysphingolipids ([PMID: 30626650](#)). Mechanistic basis: *"In mice bearing a transgene expressing the C133W SPTLC1 mutant linked to HSAN1, a 10% L-serine-enriched diet reduced dSL levels. L-serine supplementation also improved measures of motor and sensory performance"* ([PMID: 22045570](#)). NCIT: L-Serine.

Emerging — allele-specific antisense oligonucleotides (ASOs). In S331F HSN1 mice, GalNAc-conjugated LNA gapmer ASOs achieved **>90% mutant-transcript silencing** in liver, DRG, and sciatic nerve without affecting wild-type transcript, **reduced blood 1-deoxySLs**, and reversed aberrant DRG gene expression — preclinical proof of concept for allele-specific silencing ([PMID: 41124364](#)). NCIT: Antisense Oligonucleotide Therapy.

Symptomatic / supportive. - **Neuropathic pain:** standard agents (gabapentinoids, duloxetine, tricyclics, carbamazepine/oxcarbazepine for lancinating pain). NCIT: Gabapentin, Duloxetine. - **Ulcer/wound & foot care:** meticulous podiatric surveillance, protective footwear, offloading, infection control, orthopedic/surgical management of osteomyelitis and Charcot joints — the cornerstone of preventing amputations (PMID: 41161998). - **Rehabilitation:** physical/occupational therapy to preserve range of motion, strength, and balance (PMID: 41161998).

Genotype-guided strategy. Because catalytic-pocket (HSAN1A) versus transmembrane (ALS) *SPTLC1* variants produce opposite lipid signatures (1-deoxySL excess vs. canonical-SL excess), **therapy must be tailored to genotype/lipid profile**; L-serine targets the 1-deoxySL (HSAN1A) mechanism specifically (PMID: 35900868; PMID: 35904184).

Pharmacogenomics: No established PGx dosing guidance specific to HSAN1A.

13. Prevention

- **Primary prevention:** Not preventable at the population level (Mendelian). **Genetic counseling, prenatal/preimplantation genetic testing**, and reproductive options can prevent transmission. **Dietary L-serine adequacy / avoidance of alanine-loading** may reduce metabolic aggravation in carriers (mechanistically supported; PMID: 22045570).
- **Secondary prevention: Cascade genetic testing** of relatives; early biochemical (plasma 1-deoxySL) and neurophysiological monitoring to start disease-modifying therapy before advanced axon loss.
- **Tertiary prevention (most impactful): Meticulous foot care and wound surveillance** to prevent ulcers, osteomyelitis, and amputations; pain management; multidisciplinary neuromuscular care.
- **Counseling:** Autosomal-dominant recurrence risk 50%; discuss incomplete penetrance, variable expressivity, anticipation, and the possibility of the allelic ALS phenotype for specific variants.
- **Immunization / public-health / environmental interventions:** Not applicable (non-infectious, monogenic).

14. Other Species / Natural Disease

- **Orthologous gene:** *Sptlc1* in mouse (NCBI Gene ID 268656) and rat; highly evolutionarily conserved SPT machinery (yeast LCB1/LCB2 with Tsc3 small subunit).
 - **Naturally occurring disease: Hereditary sensory and autonomic neuropathies occur naturally in dogs;** a retrospective series of 11 dogs described early-onset distal sensory loss with acral mutilation and two electroclinical patterns of differing prognosis, mirroring human ulceromutilating HSAN ([PMID: 42658782](#)). (OMIA catalogs several canine sensory neuropathies.)
 - **Comparative biology/conservation:** The serine/alanine substrate-selectivity mechanism is conserved from yeast (Tsc3 regulating SPT amino-acid choice; [PMID: 30154231](#)) to humans, underpinning cross-species model validity.
 - **Zoonotic potential:** None (genetic disease).
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15. Model Organisms

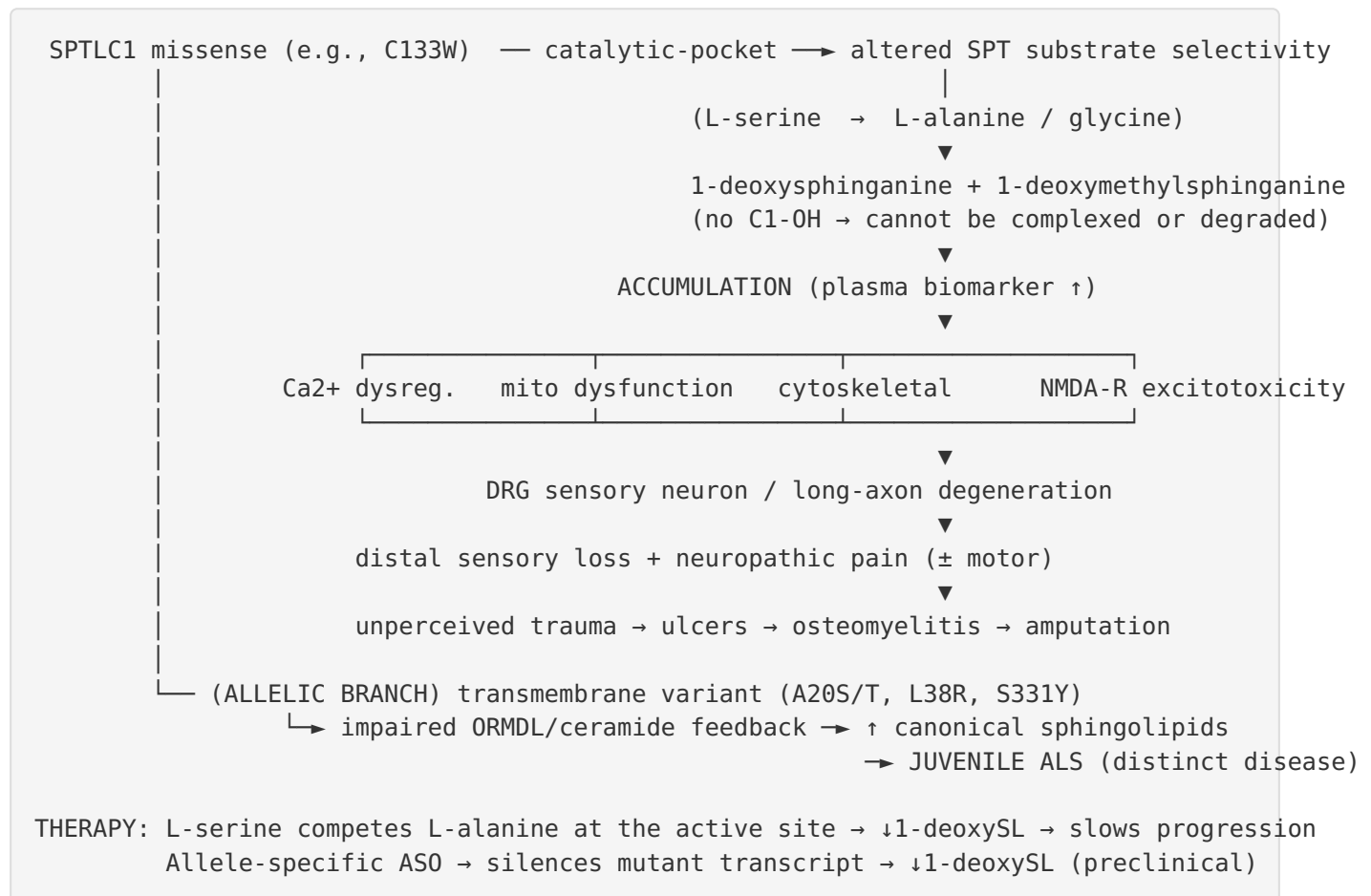
Mammalian genetic models (mouse). - **C133W *SPTLC1* transgenic mouse** — expresses the human HSAN1 mutation; accumulates deoxysphingolipids and shows sensory/motor deficits. Dietary L-serine lowered dSL and improved performance; L-alanine worsened it — the model that established the L-serine therapeutic rationale ([PMID: 22045570](#)). - **S331F *Sptlc1* knock-in mouse** — heterozygous p.S331F; elevated blood 1-deoxySLs; DRG transcriptomic (mitochondrial) changes; used for allele-specific ASO proof of concept ([PMID: 41124364](#)). - **Precision *Sptlc1* HSAN1 mouse models** engineered to recapitulate patient genetics/pathophysiology ([PMID: 34875719](#)).

Cellular / in vitro models. - **Patient iPSC-derived sensory neurons** — reveal L-serine-responsive deficits in ganglioside composition and axoglial interactions, providing a human cellular platform ([PMID: 34337561](#)). - **HEK293 / primary DRG cultures** — used to demonstrate 1-deoxySL neurotoxicity (Ca^{2+} , mitochondrial, NMDA-receptor, cytoskeletal effects) and to assay mutant SPT activity ([PMID: 29778900](#); [PMID: 27016021](#)). - **Yeast (*S. cerevisiae*)** — Tsc3/LCB system to dissect SPT amino-acid selectivity ([PMID: 30154231](#)).

Phenotype recapitulation & limitations. Mouse models reproduce the biochemical signature (elevated 1-deoxySLs) and sensory/motor deficits and respond to dSL-lowering interventions, validating mechanism and therapy. Limitations include incomplete recapitulation of the full

ulceromutilating human phenotype (chronic ulcers, osteomyelitis, amputations), and differences in disease tempo. Naturally occurring canine HSAN offers a complementary large-animal comparator ([PMID: 42658782](#)).

Mechanistic Model / Interpretation



Upstream vs downstream. The mutation and substrate shift are the **upstream** initiating lesion; 1-deoxySL accumulation is the pivotal **intermediate node** (and the therapeutic target); neuronal degeneration and ulceromutilating complications are **downstream**. The elegance of HSAN1A is that a single upstream metabolic diversion is both the biomarker source and the drug target.

A shared mechanism beyond nerve. The same 1-deoxySL/serine-deficiency axis links HSAN1A to **macular telangiectasia type 2**: "Two variants known to cause HSAN1 were identified as causal for macular telangiectasia type 2: of 11 patients with HSAN1, 9 also had

macular telangiectasia type 2. Circulating deoxysphingolipid levels were 84.2% higher among 125 patients with macular telangiectasia type 2 who did not have pathogenic variants affecting SPT than among 94 unaffected controls" (PMID: [31509666](#)). This argues HSAN1A patients may harbor sub-clinical maculopathy and broadens the therapeutic rationale for serine.

Evidence Base

PMID	Contribution	Type
11242114	Identifies <i>SPTLC1</i> (C133Y, C133W, V144D); core clinical course	Human genetics
11479835	Southern-English founder for C133W (399T>G)	Haplotype analysis
22302274	C133W most common HSAN cause in UK; screen first	Cohort/epidemiology
16364956	Reduced penetrance, anticipation, motor involvement	Clinical/path/genetic
29778900	1-deoxySL neurotoxicity: Ca ²⁺ , mitochondria; substrate-shift	In vitro
27016021	NMDA-receptor-mediated 1-deoxySL toxicity	In vitro
22045570	L-serine lowers dSL, improves mice; alanine worsens	Model organism + human
30626650	RCT: L-serine improves CMTNS by −1.5 (Class I)	Human RCT
33558762	Cryo-EM SPT–ssSPT–ORMDL3; substrate selectivity	Structural
37308477	Ceramide sensing; ALS variants impair feedback	Structural
35900868	TM-domain variants impair ORMDL → ALS lipid signature	Functional
35904184	Ser331 combines deoxySL + excess canonical SL	Functional/clinical
34459874	De novo TM variants → juvenile ALS	Human genetics
31509666	Links SPT/serine/deoxySL to MacTel-2 retina	Human + model
30154231	Tsc3/small subunit regulates serine-vs-alanine choice	Yeast
41124364	Allele-specific ASO reverses phenotype in S331F mice	Model organism
34337561	iPSC model: L-serine-responsive ganglioside deficits	In vitro (human)
26573920	<i>SPTLC2</i> HSAN1C, late-onset, raises deoxySL	Human genetics

Limitations and Knowledge Gaps

1. **Small trial size.** The pivotal L-serine RCT enrolled only 18 patients ([PMID: 30626650](#)); long-term efficacy, optimal dose, and hard clinical endpoints (ulcer/amputation rates) need larger, longer studies.
 2. **Epidemiology is imprecise.** Prevalence/incidence figures are rough; systematic registry data are lacking. The 14.3% figure is mutation-detection frequency in a referral cohort, not population prevalence.
 3. **Genotype–phenotype resolution.** Why some carriers are non-penetrant and what drives anticipation and sex-related motor severity remain unexplained; modifier genes (SPTSSA/SPTSSB) are hypothesized but not clinically validated.
 4. **Model gaps.** Rodent models capture biochemistry and mild deficits but not the full ulceromutilating phenotype; no model fully recapitulates human complications.
 5. **ASO stage.** Allele-specific silencing is preclinical; human safety/efficacy unproven ([PMID: 41124364](#)).
 6. **Retinal comorbidity.** The frequency of clinically significant MacTel-2 in HSAN1A patients and whether serine therapy protects the retina are unquantified ([PMID: 31509666](#)).
 7. **QoL data.** No disease-specific quality-of-life instrument or systematic QoL dataset exists for HSAN1A.
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Proposed Follow-up Experiments / Actions

1. **Larger, longer L-serine trial** with clinically meaningful endpoints (ulcer incidence, amputation-free survival, intraepidermal nerve-fiber density) and defined 1-deoxySL pharmacodynamic targets.
2. **First-in-human allele-specific ASO** program building on GalNAc-LNA data ([PMID: 41124364](#)), with pan-allele and mutation-specific designs.
3. **Prospective natural-history registry** capturing onset, progression rate, penetrance, sex effects, and genotype-specific outcomes to power prognostic modeling.
4. **Systematic retinal screening (OCT)** of genetically confirmed HSAN1A patients to define MacTel-2 co-occurrence and test whether serine is retino-protective.
5. **Modifier-gene study** of SPTSSA/SPTSSB and serine-metabolism genes to explain penetrance/expressivity variability and identify additional therapeutic levers.

6. **Standardized biomarker assay** for plasma 1-deoxySLs to serve as a validated companion diagnostic/pharmacodynamic endpoint across centers.
 7. **Genotype-stratified care pathway** formalizing the catalytic-pocket (HSAN1A) vs transmembrane (ALS) distinction to route patients to appropriate therapy and prognosis.
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Evidence source legend: Human clinical/genetics (family cohorts, RCT), model organism (transgenic/knock-in mouse, dog, yeast), in vitro (iPSC neurons, DRG/HEK cultures), and computational/structural (cryo-EM). Ontology suggestions provided per section (HPO, GO, CL, UBERON, CHEBI, NCIT, MONDO) for knowledge-base ingestion.

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